清華大學

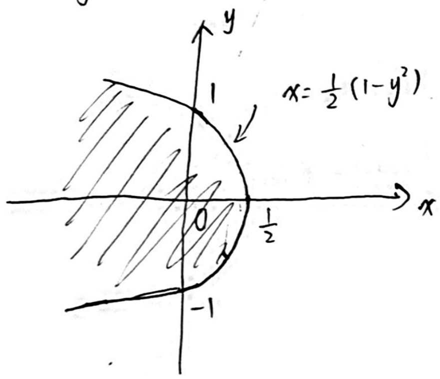
習題 1.1

(1) $|z-a|=|z-b|$

解. z 到 a、b 的距離相等，表示複平面上線段 ab 的中垂線

(2) $|z|+\mathrm{Re}\,z\le 1$

解. $\sqrt{x^2+y^2}+x\le 1 \Leftrightarrow x^2+y^2\le x^2-2x+1$ 且 $x\le 1 \Leftrightarrow x\le \frac{1}{2}(1-y^2)$ 且 $x\le 1$



上述方程所表示的平面區域是左圖的陰影部分

(3) $\mathrm{Re}\left(\frac{1}{z}\right)=2$

解. $\mathrm{Re}\left(\frac{1}{z}\right)=\frac{x}{x^2+y^2}=2 \Leftrightarrow x^2-\frac{1}{2}x+y^2=0 \Leftrightarrow \left(x-\frac{1}{4}\right)^2+y^2=\frac{1}{16}$

表示複平面上以 $\frac{1}{4}$ 為圓心，半徑為 $\frac{1}{4}$ 的圓周

(4) $\left|\frac{z-1}{z+1}\right|=2$

解. $\cancel{\left|\frac{x-1+iy}{x+1+iy}\right|=2 \Leftrightarrow \frac{|x^2-y^2-1|}{(x+1)^2+y^2}}$

$|z-1|=2|z+1| \Leftrightarrow (x-1)^2+y^2=4\left[(x+1)^2+y^2\right]$

$\Leftrightarrow (3x+1)(x+3)+3y^2=0$

$\Leftrightarrow 3x^2+10x+3y^2+3=0$

$\Leftrightarrow \left(x+\frac{5}{3}\right)^2+y^2=\left(\frac{4}{3}\right)^2$

表示複平面上以 $-\frac{5}{3}$ 為圓心，半徑為 $\frac{4}{3}$ 的圓周

(5) $\arg(z-i)=\frac{\pi}{4}$

解. $\Leftrightarrow \frac{y-1}{x}=1$ 且 $x>0 \Leftrightarrow y=x+1$ 且 $x>0 \Leftrightarrow \frac{1}{2i}(z-\bar{z})=\frac{1}{2}(z+\bar{z})+1 \Leftrightarrow \left(\frac{1}{2}+\frac{i}{2}\right)z+\left(\frac{1}{2}-\frac{i}{2}\right)\bar{z}+1=0$

表示複平面上 ~~y=x+1~~ $(1+i)z+(1-i)\bar{z}+2=0$ 的直線

習題 1.2

(1) $(1+i)^i$

解. 原式 $= (\sqrt{2}e^{i\frac{\pi}{4}})^i = e^{-\frac{\pi}{4}+i\cdot\ln\sqrt{2}} = e^{-\frac{\pi}{4}}\left(\cos\left(\frac{1}{2}\ln 2\right)+i\sin\left(\frac{1}{2}\ln 2\right)\right)$

(2) $\cos\varphi+\cos 2\varphi+\dots+\cos n\varphi$

解：設 $z=\cos\varphi+i\sin\varphi$，則原式 $= Re\left(\sum_{k=1}^{n} z^k\right) = Re\left(\frac{z-z^{n+1}}{1-z}\right)$

$$=\frac{1}{2}\left(\frac{z-z^{n+1}}{1-z}+\frac{\bar z-\bar z^{n+1}}{1-\bar z}\right) = \frac{1}{2}\cdot\frac{(z+\bar z)-2z\bar z-(z^{n+1}+\bar z^{n+1})+(z^{n+1}\bar z+\bar z^{n+1}z)}{1-(z+\bar z)+z\bar z}$$

$$=\frac{1}{2}\cdot\frac{2\cos\varphi-2-2\cos(n+1)\varphi+2\cos n\varphi}{2-2\cos\varphi} = \frac{1}{2}\cdot\frac{\cos\varphi+\cos n\varphi-\cos(n+1)\varphi-1}{1-\cos\varphi}$$

(3) $\sin(a+ib)$

解. ~~原式 $= Im(e^{i(a+ib)}) = Im(e^{-b+ia}) = e^{-b}\sin a$~~

原式 $= \frac{1}{2i}\left(e^{i(a+ib)}-e^{-i(a+ib)}\right) = \frac{1}{2i}\left[e^{-b}(\cos a+i\sin a)-e^{b}(\cos a-i\sin a)\right]$

$$=\frac{(e^b+e^{-b})}{2}\sin a+i\frac{e^b-e^{-b}}{2}\cos a$$

習題 1.3

證：$\overline{e^z} = \overline{e^{x+iy}} = e^x\cdot\overline{e^{iy}} = e^x\cdot e^{-iy} = e^{x-iy} = e^{\bar z}$

習題 1.4

(1) 其參數方程是 $z=(1+3i)t+(-1+4i)(1-t) = (2t-1)+(4-t)i$

$$t=\frac{\frac{1}{2}(z+\bar z)+1}{2} = 4-\frac{1}{2i}(z-\bar z) \Leftrightarrow \left(\frac{1}{4}-\frac{i}{2}\right)z+\left(\frac{1}{4}+\frac{i}{2}\right)\bar z-\frac{7}{2}=0$$

$$\Leftrightarrow (1-2i)z+(1+2i)\bar z-14=0$$

(2) 設圓心是 z_0，半徑是 r，則有 $|z-z_0|=r \Leftrightarrow zz^*-zz_0^*-z^*z_0+z_0z_0^*=r^2$

$$\Leftrightarrow Azz^*-Az_0z^*-Az_0^*z+A(z_0z_0^*-r^2)=0 \quad (A\neq 0)$$

所以有 $B=-Az_0$，$C=A(z_0z_0^*-r^2) \Leftrightarrow z_0=-\frac{B}{A}$，$C=A\left(\frac{B\bar B}{A\bar A}-r^2\right)$

$$\Rightarrow r^2=\frac{BB^*}{AA^*}-\frac{C}{A}>0 \Leftrightarrow BB^*-CA>0 \Leftrightarrow |B|^2>CA$$

因此當 $A\neq 0$ 且 $|B|^2>CA$ 時，上述方程才表示為圓

圓心是 $-\frac{B}{A}$，半徑是 $\frac{1}{|A|}\sqrt{BB^*-CA}$

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习题 1.5

$$f'(z) = \lim_{\Delta r\to 0} \frac{u(r+\Delta r,\theta)+iv(r+\Delta r,\theta) - u(r,\theta) - iv(r,\theta)}{(r+\Delta r)e^{i\theta} - re^{i\theta}}$$

$$= \left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) e^{-i\theta} = \frac{r}{z}\left(\frac{\partial u}{\partial r} + i\frac{\partial v}{\partial r}\right) \quad \text{沿径向求导}$$

$$f'(z) = \lim_{\Delta\theta\to 0} \frac{u(r,\theta+\Delta\theta)+iv(r,\theta+\Delta\theta) - u(r,\theta) - iv(r,\theta)}{re^{i(\theta+\Delta\theta)} - re^{i\theta}}$$

$$= \left(\frac{\partial u}{\partial \theta} + i\frac{\partial v}{\partial \theta}\right)\cdot\frac{1}{ire^{i\theta}} = \left(\frac{1}{r}\frac{\partial v}{\partial \theta} - i\frac{1}{r}\cdot\frac{\partial u}{\partial \theta}\right) e^{-i\theta} \quad \text{沿圆周求导}$$

$$\Rightarrow \begin{cases} \dfrac{\partial u}{\partial r} = \dfrac{1}{r}\dfrac{\partial v}{\partial \theta} \\ \dfrac{\partial v}{\partial r} = -\dfrac{1}{r}\dfrac{\partial u}{\partial \theta} \end{cases} \quad \text{C-R条件}$$

习题 1.6

(1) $\frac{\partial u}{\partial x} = y^2$, $\frac{\partial v}{\partial y} = x^2$, $\frac{\partial u}{\partial y} = 2xy$, $\frac{\partial v}{\partial x} = 2xy$

要满足C-R条件，(x,y)只能是$(0,0)$

即只在原点可微，因在複平面上處處不解析

(2) $\frac{\partial u}{\partial x} = 6x^2$, $\frac{\partial v}{\partial y} = 9y^2$, $\frac{\partial u}{\partial y} = 0 = \frac{\partial v}{\partial x}$

只在 $\sqrt{6}x-3y=0$ 和 $\sqrt{6}x+3y=0$ 两条直线上可微，在複平面上處處不解析

习题 1.7

(1) $$v = \int^{(x,y)} -\frac{\partial u}{\partial y}dx + \frac{\partial u}{\partial x}dy = \int^{(x,y)} -(e^x\cos y)dx + (e^x\sin y)dy = \int^{(x,y)} d(-e^x\cos y) = -e^x\cos y + C$$

$$\cancel{= -e^x\cos y - e^x\cos y + C = -2e^x\cos y + C}$$

$$f(z) = e^x\sin y - ie^x\cos y + Ci$$

(2) $u = \dfrac{r^2(\cos^2\theta - \sin^2\theta)}{r^4} = \dfrac{\cos 2\theta}{r^2}$

$$v = \int^{(r,\theta)} r\frac{\partial u}{\partial r}d\theta + \left(-\frac{1}{r}\right)\cdot\frac{\partial u}{\partial\theta}dr = \int^{(r,\theta)} r\cdot\left(-2\cdot\frac{\cos 2\theta}{r^3}\right)d\theta - \frac{1}{r}\cdot\frac{-2\sin 2\theta}{r^2}dr$$

$$= -\int^{(r,\theta)} d\left(\frac{\sin 2\theta}{r^2}\right) = -\frac{\sin 2\theta}{r^2} + C$$

$$0 = \lim_{r\to\infty}\left(\frac{\cos 2\theta}{r^2} + i\left(-\frac{\sin 2\theta}{r^2} + C\right)\right) = Ci \Rightarrow C = 0.$$

$$\Rightarrow f(z) = \frac{\cos 2\theta}{r^2} - i\cdot\frac{\sin 2\theta}{r^2} = \frac{1}{r^2}\cdot e^{-i\cdot 2\theta} = \frac{1}{z^2}$$

(3) $v = \displaystyle\int^{(\rho,\theta)} \rho\cdot\frac{\partial u}{\partial\rho}d\theta + \left(-\frac{1}{\rho}\cdot\frac{\partial u}{\partial\theta}\right)d\rho = \int^{(\rho,\theta)} d\theta = \theta + C$

$0 = f(1) = \ln 1 + (0 + C)i \Rightarrow C = 0$

$\Rightarrow f(z) = \ln\rho + i\theta = \ln z$